

SHETH LUJ COLLEGE

Decision Tree Learning

- 1 Implement the Decision Tree Learning algorithm to build a decision tree for a given dataset.
- 2 Evaluate the accuracy and effectiveness of the decision tree on test data.
- 3 Visualize and interpret the generated decision tree.

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Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Daniyal khan/AI_PRACTICAL/decesion_tree.py =====
Predicted Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]

Actual Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]

Accuracy: 100.0 %

Confusion Matrix:
[[10 0 0]
 [0 9 0]
 [0 0 11]]

Classification Report:
      precision    recall  f1-score   support

 setosa      1.00      1.00      1.00        10
 versicolor  1.00      1.00      1.00         9
 virginica    1.00      1.00      1.00        11

 accuracy      1.00      1.00      1.00        30
 macro avg     1.00      1.00      1.00        30
 weighted avg   1.00      1.00      1.00        30
```

